



BD- Apresentação Final CoinCollector

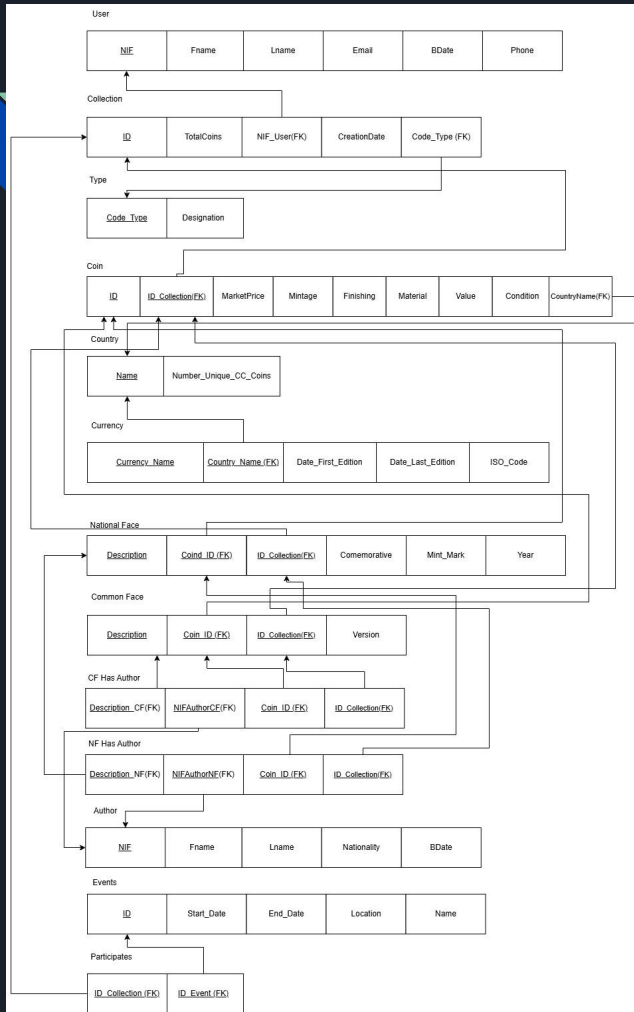
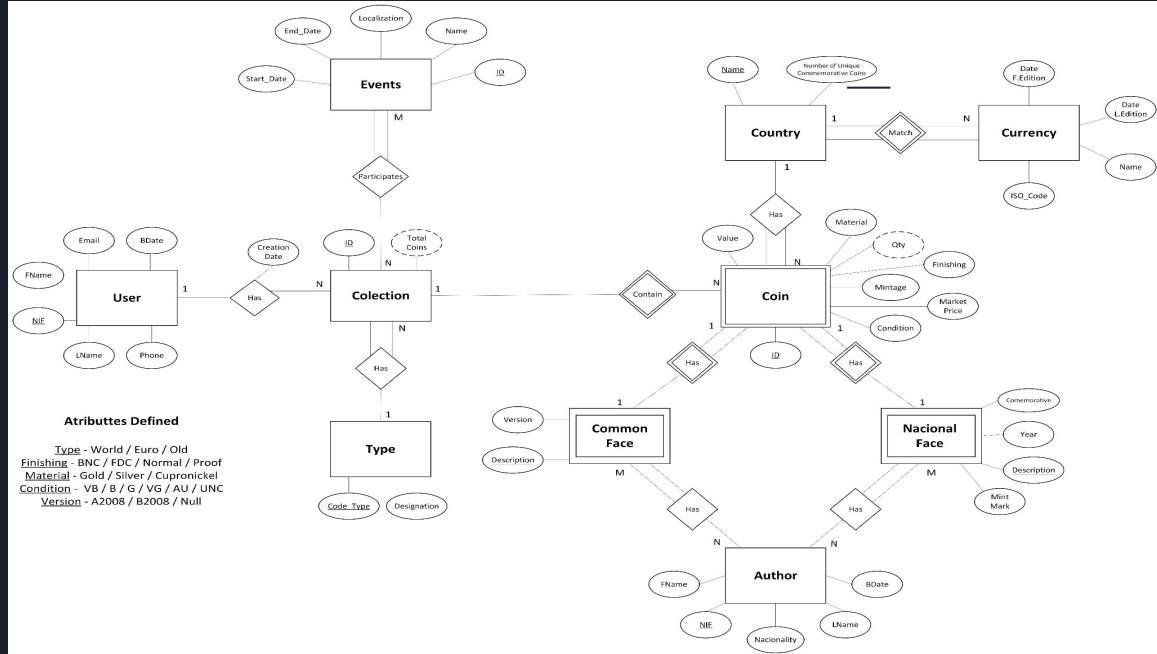
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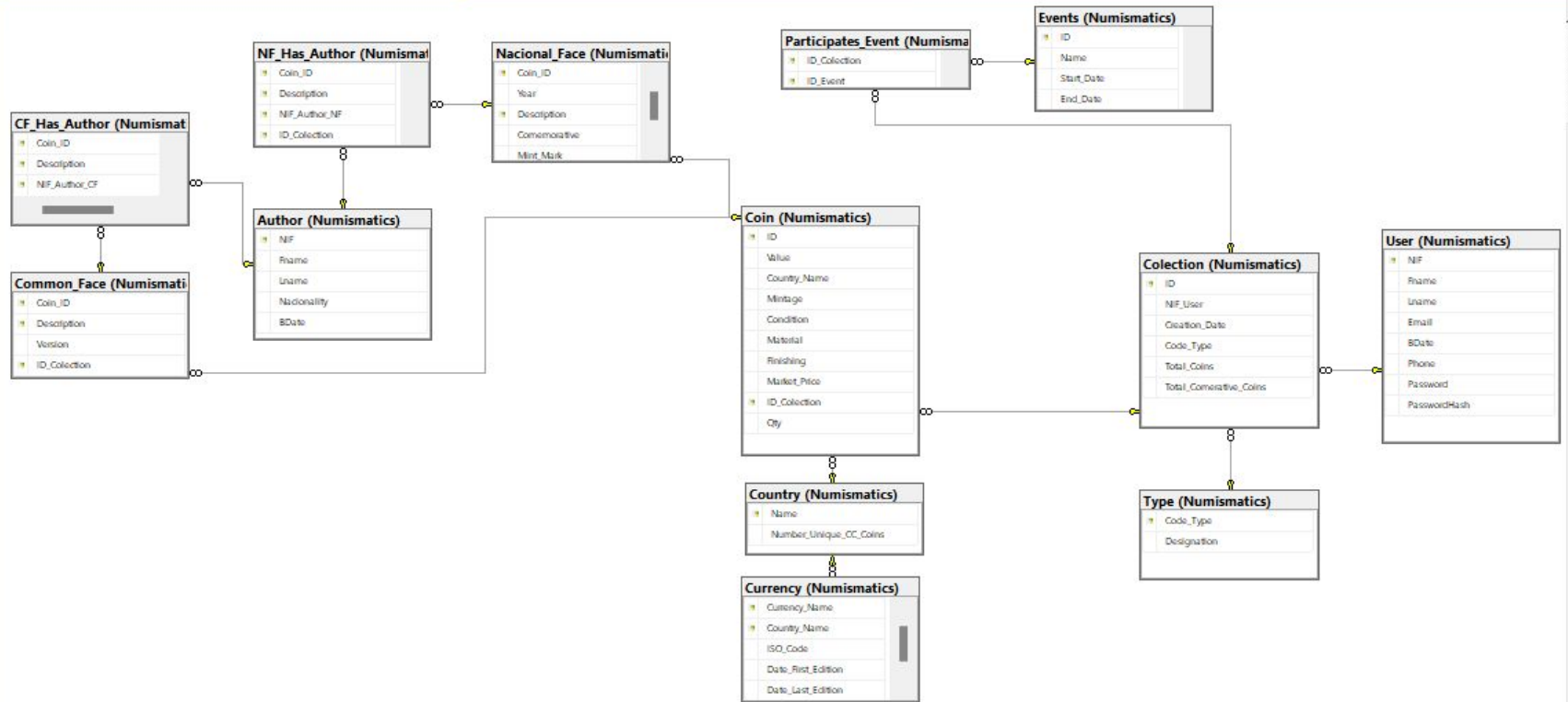
Introdução

- Aplicação web para gestão de coleções de moedas
- Desenvolvida com Flask (Python) e SQL Server
- Funcionalidades Principais:
 - Autenticação e perfis de utilizadores
 - Inventário de moedas
 - Gestão de coleções
 - Participação em eventos
 - Gestão de autores
- Destinada a colecionadores de moedas

ER/DER



Modelo Relacional





Elementos Importantes

Stored
Procedures

39

Sql Security

Indexes

16

UDF

5

Triggers

5

Transactions

6

```

ALTER PROCEDURE [Numismatics].[AddEventWithParticipation]
    @EventName VARCHAR(25),
    @StartDate VARCHAR(10),
    @EndDate VARCHAR(10),
    @Localization VARCHAR(40),
    @UserNIF INT,
    @NewEventID INT OUTPUT
AS
BEGIN
    SET NOCOUNT ON;

    BEGIN TRY
        BEGIN TRANSACTION;

        -- Get the next available Event ID
        SELECT @NewEventID = ISNULL(MAX(ID), 0) + 1 FROM [Numismatics].[Events];

        -- Insert the new event
        INSERT INTO [Numismatics].[Events] (ID, [Name], [Start_Date], End_Date, Localization)
        VALUES (@NewEventID, @EventName, @StartDate, @EndDate, @Localization);

        -- Get all collections of the user
        INSERT INTO [Numismatics].Participates_Event (ID_Collection, ID_Event)
        SELECT ID, @NewEventID
        FROM [Numismatics].Collection
        WHERE NIF_User = @UserNIF;

        COMMIT TRANSACTION;
    END TRY
    BEGIN CATCH
        IF @@TRANCOUNT > 0
            ROLLBACK TRANSACTION;

        THROW;
    END CATCH
END

```

Stored Procedures

- Adicionar / Remover / Editar Moedas
- Adicionar / Remover Coleções
- Adicionar Eventos
- Set / Validate User Password
- Alguns Listing e Getters

```
CREATE OR ALTER TRIGGER tr_AfterInsertNacionalFace
ON Nacional_Face
AFTER INSERT
AS
BEGIN
    SET NOCOUNT ON;

    UPDATE c
    SET Total_Comerative_Coins = Total_Comerative_Coins + 1
    FROM Colection c
    JOIN inserted i ON c.ID = i.ID_Colection
    WHERE i.Comemorativa = 1;

    PRINT ('TRIGGER: Moeda comemorativa adicionada');
END;
```

Triggers

Tabela Coin

- After Insert Coin
- After Delete Coin

Table Nacional_Face

- After Delete NF
- After Insert NF

Tabela Colection

- Delete Coins From Colection

```

CREATE FUNCTION fn_GetEventParticipation (
    @CollectionID INT,
    @OnlyFuture BIT
)
RETURNS @EventTable TABLE (
    EventID INT,
    EventName VARCHAR(25),
    Start_Date VARCHAR(10),
    End_Date VARCHAR(10),
    Localization VARCHAR(40)
)
AS
BEGIN
    IF @OnlyFuture = 1
    BEGIN
        INSERT INTO @EventTable
        SELECT
            E.ID,
            E.Name,
            E.Start_Date,
            E.End_Date,
            E.Localization
        FROM [Numismatics].Events E
        INNER JOIN Participates_Event PE
            ON PE.ID_Event = E.ID
        WHERE
            PE.ID_Collection = @CollectionID
            AND TRY_CAST(E.End_Date AS DATE) >= CAST(GETDATE() AS DATE)
    END
    ELSE
    BEGIN
        INSERT INTO @EventTable
        SELECT
            E.ID,
            E.Name,
            E.Start_Date,
            E.End_Date,
            E.Localization
        FROM [Numismatics].Events E
        INNER JOIN Participates_Event PE
            ON PE.ID_Event = E.ID
        WHERE PE.ID_Collection = @CollectionID
    END
    RETURN
END
GO

```

User Defined Functions

Multi-statement Table-Valued

- GetEventParticipation

Escalares

- GetCollectionValue
- GetAuthorAge
- GetUserFullName

Inline Table-valued

- GetAuthorsOfCollection



DEMO

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Conclusão e Trabalho Futuro

- Este projeto ajudou a consolidar conhecimentos sobre bases de dados, da análise à implementação.
- Trabalho Futuro :
 - Roles
 - Analisar os índices para melhorar o desempenho da base de dados
 - Transpor alguns Stored Procedures para UDFs, nomeadamente os getters.